

Forward and inverse problems in causal discovery

IA pour la Science

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This document explains the scheme of causal structure selection and generation

The empirical distribution of the observations

$P(X_1, \dots, X_n)$ is the distribution of the observable data. Its approximation is given by the Gaussian Process Regression model

$$p(\mathbf{x} \mid f_{\text{gp}})$$

on the dataset

$$\{\mathbf{x}_i \in \mathbb{R}^n\}.$$

It is indexed by some function of the time and space $(t, s) \mapsto i$.

A model from the family of universal approximators

For a sample $\mathbf{x}$ and its reconstruction $\hat{\mathbf{x}}$ the likelihood function provided a universal model $f_{\text{nn}}(\boldsymbol{\theta}, \mathbf{x})$ is

$$p(\mathbf{x} \mid f_{\text{nn}}, \boldsymbol{\theta}).$$

Without the a priori information about the model parameters $\boldsymbol{\theta}$ the corresponding loss function is

$$\mathcal{L}(\boldsymbol{\theta}) = -\log p(\mathbf{x} \mid f_{\text{nn}}, \boldsymbol{\theta}) = (\hat{\mathbf{x}} - \mathbf{x})^\top \mathbf{B}^{-1}(\hat{\mathbf{x}} - \mathbf{x}).$$

The matrix $\mathbf{B}$ is the covariance of the reconstruction error. This matrix in some states depends on the index i . This model as a parametric one has a structure to make a distillation to the causal graph

$$f_{\text{nn}}(\boldsymbol{\theta}, \mathbf{x}) \xrightarrow{\text{extract}} \mathbf{Z}_{\text{nn}} \xrightarrow{\text{dist}} \mathbf{Z}.$$

An SCM we have to forecast

A structural causal model

$$\hat{\mathbf{x}} = f_G(\mathbf{w}, \mathbf{x}) \in \mathcal{F}_{\text{scn}}$$

has a structure $\mathbf{Z}$ and its probabilistic representation $\mathbf{\Gamma}$. The elements of the matrix $\mathbf{\Gamma}$ are the probabilities of the causal relationships between the elements of the model structure f_G . The matrix $\mathbf{Z}$ is the adjacency matrix of the causal graph, the binary representation of $\mathbf{\Gamma}$.

Note: we suppose that SCM does not fit the whole (s, t) -data but a local part of it. So we also call SCM a local model f_G , in contrast to the universal model f_{nn} .

A foundational model to forecast the SCM

There given pairs a sample $\mathbf{x}_i$ and the corresponding SCM f_G with its structure $\{\mathbf{Z}_i\}$. Call the set of pairs

$$(\mathbf{x}_i, \mathbf{Z}_i), \quad i \in \mathcal{B}$$

a collection of the SCM forecasting. This collection $\mathcal{B}$ shows one case of mathematical modeling that results in an interpretable model f_G . This model fits samples $\mathbf{x}_i$ from some local data $i \in \mathcal{B}$.

Call a model that forecasts an SCM $\mathcal{F}$ a *foundational model*

$$f_{ai} : \mathbf{x}_B \mapsto f_G \in \mathcal{F}.$$

The SCM is a parametric model

$$f_G = f_G(\mathbf{w}, \mathbf{x}) \quad \text{with the parameters} \quad \hat{\mathbf{w}} = \arg \min_{\mathbf{w}} \mathcal{L}(\mathbf{w}).$$

The foundational model returns not the SCM $f_G(\hat{\mathbf{w}}, \mathbf{x})$ itself, but only a probabilities $\mathbf{\Gamma}$ on the structure $\mathbf{Z}$. So we rewrite the previous definition as

$$f_{ai} : \mathbf{x}_B \mapsto \mathbf{\Gamma}.$$

Note: Since a case of mathematical modeling, call it $\mathcal{B}$ has a history of requests to the foundational model, the next information is available for it:

- 1) the optimal parameters $\mathbf{w}$ along with $\mathbf{Z}, \mathbf{\Gamma}$ for the SCM f_G ,
- 2) the prior probabilities of the model parameters $p(f_G)$ or all models in the family $\mathcal{F}$,
- 3) the other information about the case of mathematical modeling $\mathcal{B}$ that is relevant for the forecasting of the SCM.

The Integer Linear Programming for the causal graph selection

The matrix $\mathbf{\Gamma}$ comes along with the descriptions of its nodes $X_1, \dots, X_n$, and the probabilities. The SCM f_G is defined by the graph $G = (\mathbf{Z}, X)$ by solving

$$\text{ILP}(\mathbf{\Gamma}, P_X) \rightarrow \mathbf{Z} \rightarrow f_G.$$

The evidence of the SCM

A probabilistic description Γ of a model f_G from the class of *admissible models* $\mathcal{F}$ is generated from a parametric distribution

$$\Gamma \sim \text{Dir}(\boldsymbol{\pi}) \quad \text{generated by MCMC} \quad f_{\text{mh}}(\boldsymbol{\pi}),$$

here the lowercase letters mean Metropolis-Hastings Algorithm for MCMC. (A simple genetic algorithm may bring some decent results, too). This distribution provides a probabilistic space for the similar models. See the document on *Multimodeling*.

The model f_G and the corresponding graph has its likelihood

$$p(\mathbf{x}|f_G).$$

See the document on *Causal graph likelihood*.

Note: *Two-level bayesian inference introduces the probability of the SCM parameters on the first level. And the probability of the model on the second level is:*

- 1) *the prior of the SCM is delivered by the generative model f_{mh}*
- 2) *the likelihood of the SCM is delivered by the forecasting model f_{ai} .*

The criterion of causality

The parameters of the model f_G has their covariance matrix. The prior of the model parameters is

$$-\ln p(\mathbf{w}|f_g) = (\mathbf{w} - \boldsymbol{\mu})^T \mathbf{A}^{-1} (\mathbf{w} - \boldsymbol{\mu}).$$

The algorithm of the Bayesian hyperparameters $(\boldsymbol{\mu}, \mathbf{A})$ estimation (and their connection between the causality) is described in the document on *parameter analysis*.

A model for the intervention

See the document on the *intervention strategy*.